

NET Centric Computing Project

FIFA Squad Builder

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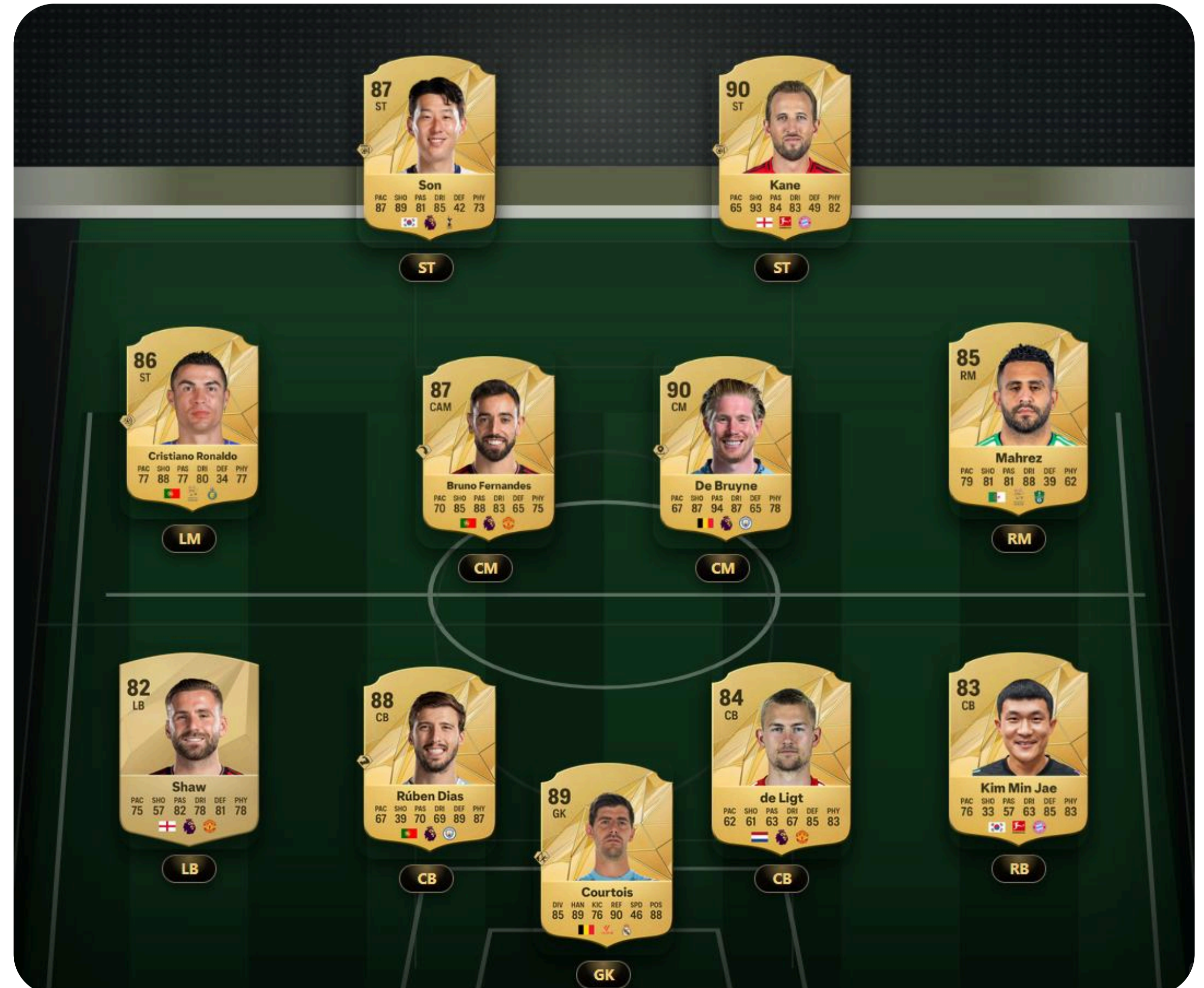
Tools Used:

ASP.NET Core MVC, C#, HTML, CSS, MySQL



FIFA Squad Builder

- A full-stack web app for building FIFA Ultimate Team-style football squads
- Powered by a real FIFA 23 player dataset (~18,500 players, 679 clubs)
- Lets users pick real players, arrange them on a formation, and see live squad stats — rating, chemistry, age, value, wages
- Built with ASP.NET Core MVC, C#, Entity Framework Core, and MySQL



Problem Statement

- FIFA Ultimate Team is closed — no way to inspect or learn from it
- Real player datasets exist, but only as flat CSVs
- Gap: turning raw data into a working, secure, multi-user squad builder



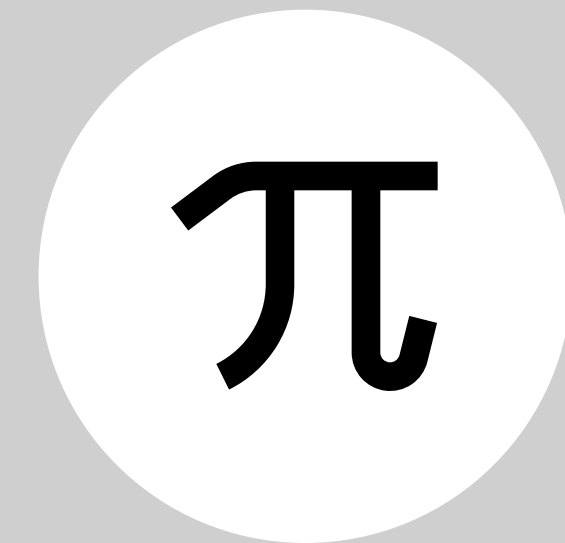
Objectives



Real relational DB from
real FIFA23 dataset (18.5k
players)

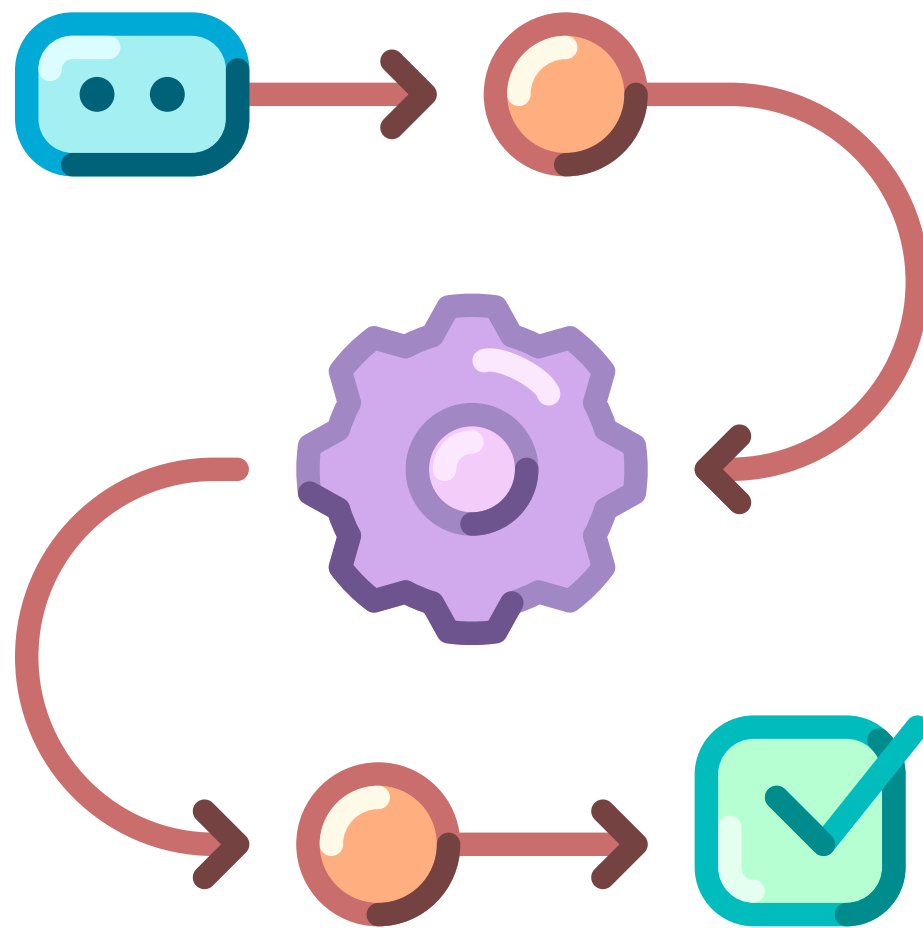


Secure, role-based, multi-
user squad builder
(ownership enforced
server-side)

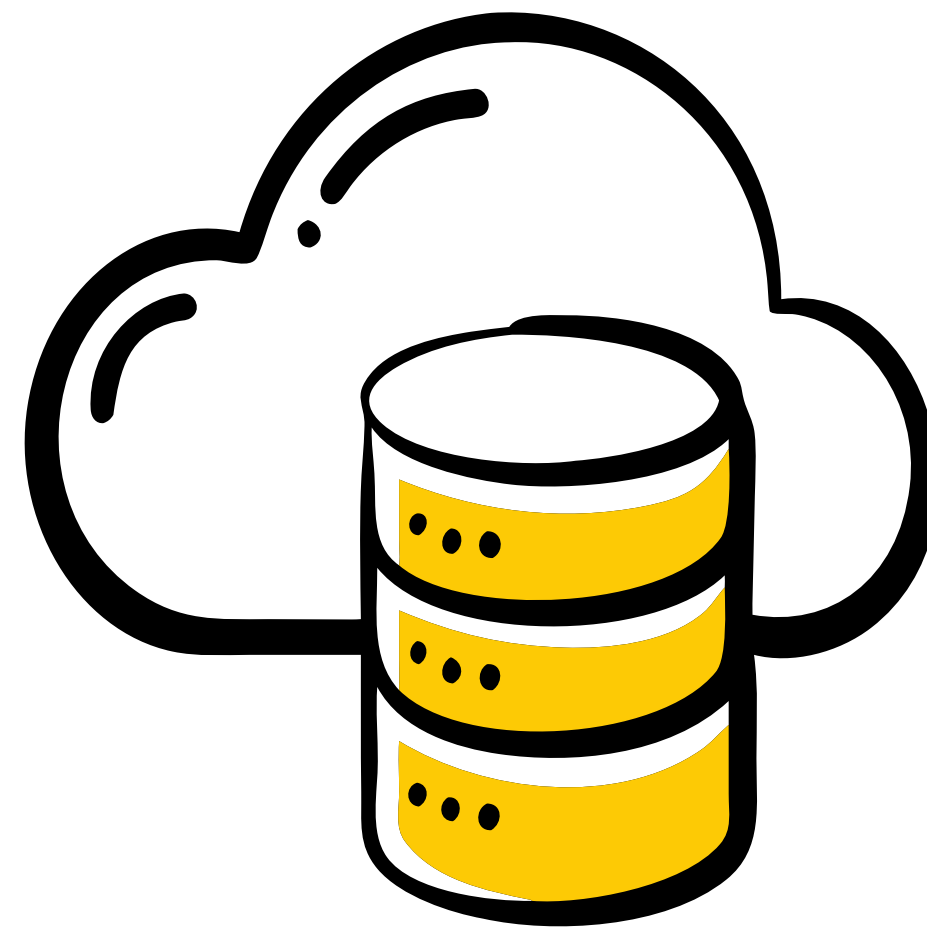


Original, testable
chemistry/rating/weak-
position algorithms

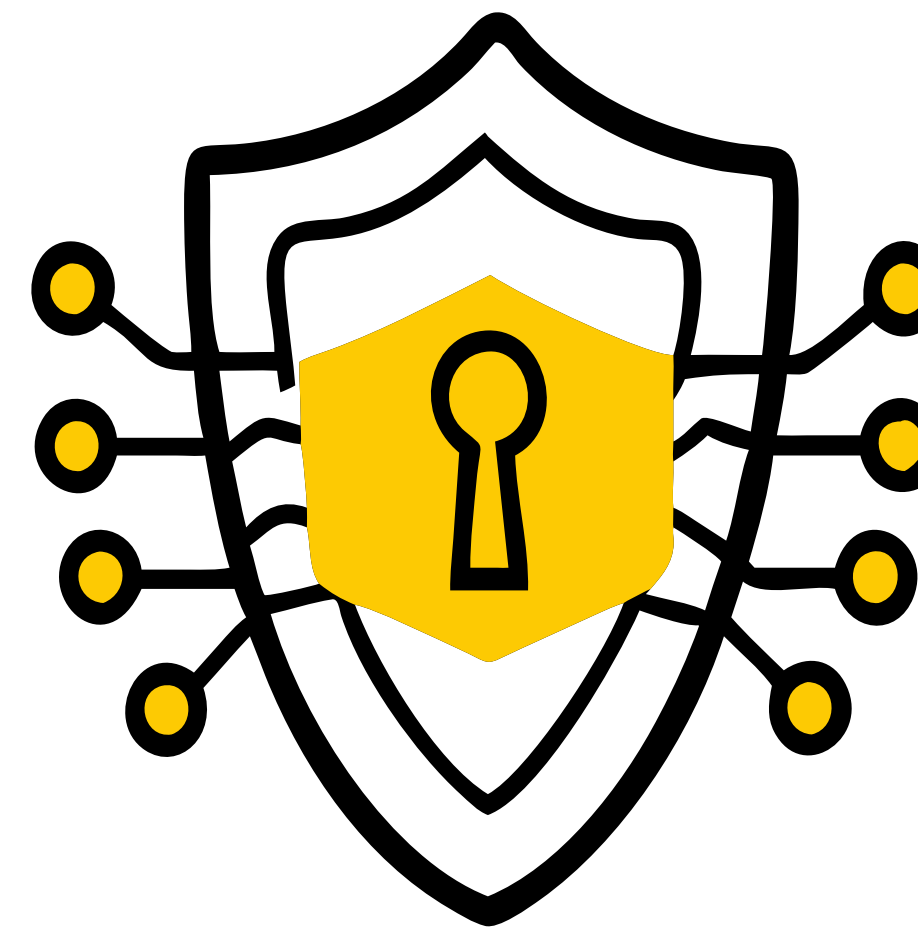
Tech Stack



ASP.NET Core
MVC (.NET 8 LTS),
C#



EF Core 8 +
MySQL (Pomelo
provider)



ASP.NET Core
Identity (roles:
User, Admin)



xUnit + EF Core
InMemory (21
tests)

Features



Squad & Formation Building

Handles squad creation, interactive pitch layouts, and player assignments

Admin Administration

Provides CRUD management for players, formations, and user roles

Data Management & Import

Imports and queries a dataset of ~18,500 FIFA23 players idempotently

Positional Optimization

Identifies weak positions by comparing squad slots against top available players.

Authentication & Access Control

Secures the system using ASP.NET Core Identity with User and Admin roles

Data Integrity Enforcement

Prevents duplicate players and slot conflicts directly at the database level

Real-Time Analytics

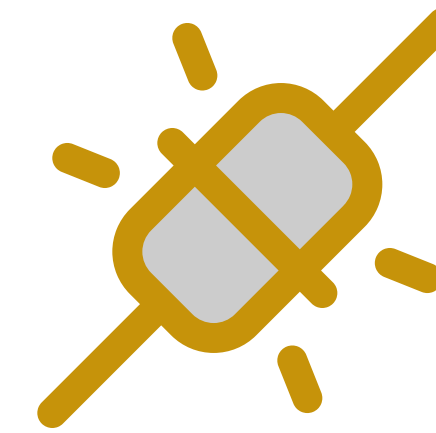
Tracks live chemistry, ratings, age, potential, squad value, and wage budgets

Modules



Squad Builder

Formation-driven pitch (DB-stored X/Y coordinates)



API Module

Frontend & backend connection



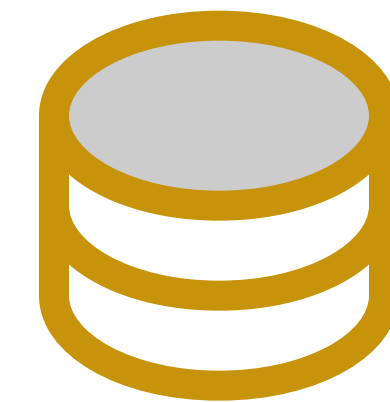
Admin Panel

Player/Formation CRUD
Dataset import trigger
User role management



Authentication

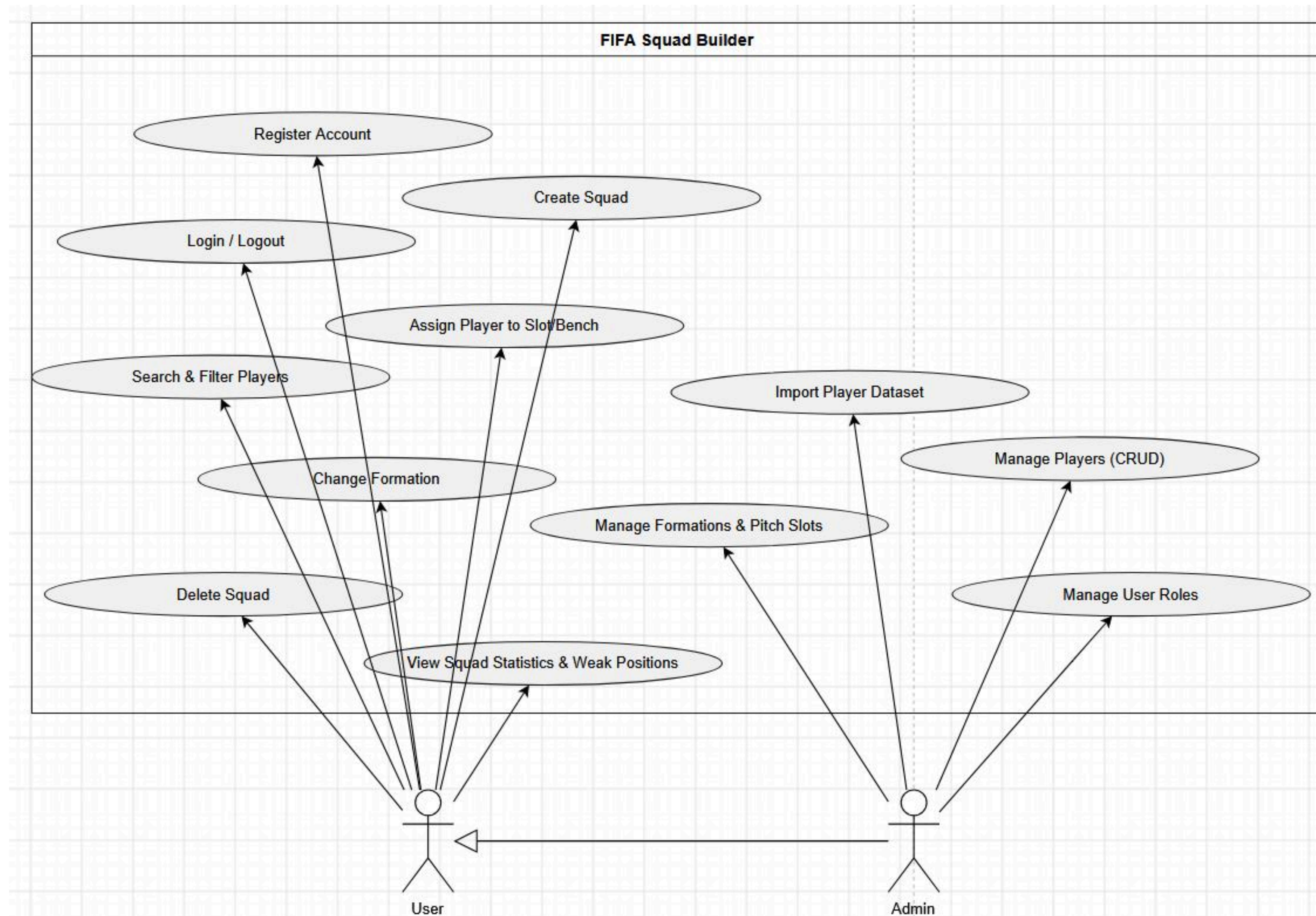
User login
Access control



Calculations

Chemistry
Squad rating
Weak positions

Use Case Diagram



Future Implementation

- Drag-and-drop player placement (currently click-to-search)
- Social features: share or compare squads between users
- Per-position chemistry heatmap / visual breakdown
- Save/checkpoint-style squad versioning
- Public deployment with HTTPS



Key Takeaways

- Working with a real, messy dataset forced actual data-validation skills, not assumptions
- MVC enables separation of concerns and parallel development
- Honest reporting of limitations (build/test execution constraints) is as important as the working features themselves
- User-friendly interface
- Optimizing indexing



